Acceleration Investigation

1. PLANNING

1.1 Aim

What are you trying to find out? What problem are you trying to solve?

_____

_____

1.2 Variables

You can only change one variable at a time and measure its effect. The rest of the variables must be kept the same.

Change (Independent)	Measure (Dependent)	Keep the Same (Controlled)
_____	_____	_____
		_____
		_____

1.3 Hypothesis

A statement about what you think will happen. This statement needs to link the independent and dependent variables.

_____

_____

2. CONDUCTING AND RECORDING

2.1 Equipment

Write a list of all the equipment you will need to conduct your experiment.

- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____

2.2 Procedure

Design a method by which you can slowly change one variable and measure its affect on the other.

1. _____

   _____

2. _____

   _____

3. _____
_____
4. _____
_____
5. _____
_____
6. _____
_____
7. _____
_____
8. _____
_____
9. _____
_____
10. _____
_____
___

2.3 Diagram of Equipment Set-up

2.4 Results

Using your ticker timer tape, complete the results tables below.

Angle 1:

Interval	Distance (cm)	Average Speed (cm/s)	Change in Speed (cm/s)	Acceleration (cm/s^2)
1				
2				
3				
4				
5				
6				
		Average Acceleration		

Angle 2:

Interval	Distance (cm)	Average Speed (cm/s)	Change in Speed (cm/s)	Acceleration (cm/s^2)
1				
2				
3				
4				
5				
6				
		Average Acceleration		

Angle 3:

Interval	Distance (cm)	Average Speed (cm/s)	Change in Speed (cm/s)	Acceleration (cm/s^2)
1				
2				
3				
4				
5				
6				

	Average Acceleration	

2.5 Class Results

Draw a table in the space below to record the class results. You will need two columns, one for angle of slope and one for acceleration.

3. ANALYSING

3.1 Graphing

Draw a scatter plot of the class results. The angle of the slope should be on the x-axis and the average acceleration should be on the y-axis. Draw a **line of best fit** through the points on your graph.

Glue your graph on the back page.

3.2 Analysing Your Results

Do your results support your hypothesis or not?

_____

_____

Explain **how** you used your results to come to your conclusion above.

_____

_____

_____

_____

_____

4. EVALUATION

4.1 Errors

What are the possible sources of error in your investigation?

_____

_____

_____

_____

How might these sources of error affect your data?

_____

_____

_____

_____

4.2 Improving Your Investigation

What changes would you make to your investigation to improve the accuracy of your results?

_____

_____

_____

_____

_____